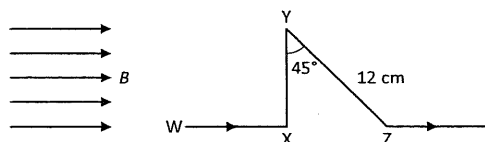


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/23]

A current of 5.0 A is flowing in the wire as shown in the diagram below.



Given that the magnetic flux density is $B = 0.40 \text{ T}$, what is the force on the portion YZ of the wire?

	Force	Direction
A	0.17 N	Into the page
B	0.17 N	Out of the page
C	0.24 N	Into the page
D	0.24 N	Out of the page

2. [ACJC/H2/Prelims 2018/P1/24]

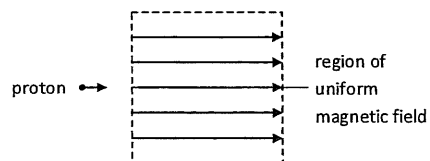
A particle X of charge $2q$ is moving in a uniform magnetic field of flux density B in a circle of radius r at a speed v . Another particle Y of charge q has the same mass as X and moves at the same speed as X.

What is the flux density required for Y to move in a circle which has half the radius of the circle moved by X?

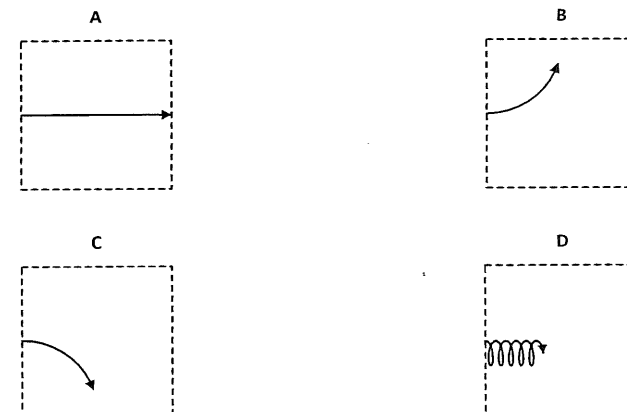
- A $\frac{B}{4}$ B $\frac{B}{2}$ C $2B$ D $4B$

3. [AJC/H2/Prelims 2018/P1/22]

A proton enters a region of uniform magnetic field. The direction of the particle's velocity is parallel to the direction of the magnetic field as shown in the diagram below.

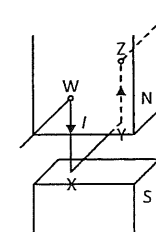


Which of the following diagrams correctly shows the path of the proton while in the region of magnetic field?



4. [JC/H2/Prelims 2018/P1/25]

A piece of wire WXYZ is pivoted freely about a horizontal axis at points W and Z. Section XY of the wire is situated between the North (N) and South (S) poles of a horse-shoe magnet. WXYZ is connected to an electrical circuit.

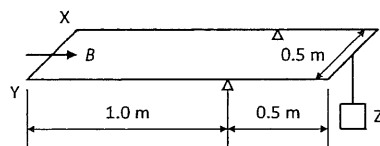


What will happen to the wire, if any, when the circuit is just turned on and there is a constant current I in the wire as shown?

- A Swings to the left.
 B Swings to the right.
 C Swings from X to Y.
 D Remains at rest at its original position.

5. [DHS/H2/Prelims 2018/P1/23]

A 1.5 m by 0.5 m light and rigid rectangular conducting frame is pivoted along its longer sides with a weight Z hung on one shorter side as shown. A uniform horizontal magnetic field B of flux density 0.050 T is applied at right-angles to the section XY of the frame.

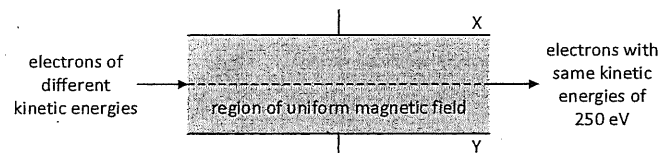


When a current passes through the section XY of the frame, which combination of the magnitude and direction of current flowing in section XY , and the weight Z makes the frame horizontal?

	magnitude of current in section XY	direction of current in section XY	Z / N
A	1.96 A	from X to Y	0.049
B	1.96 A	from Y to X	0.098
C	3.92 A	from X to Y	0.196
D	3.92 A	from Y to X	0.098

6. [DHS/H2/Prelims 2018/P1/24]

In certain experiments involving scattering of electrons by nucleus, a beam of electrons of kinetic energy 250 eV are needed. It can be obtained by passing a beam of electrons of different kinetic energies through a velocity selector as shown, with plate Y at a higher potential with respect to plate X .

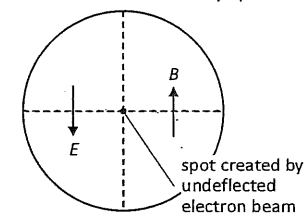


Which of the following gives the correct effect on electrons that enter the velocity selector with kinetic energies that differs from the required 250 eV?

	electrons with kinetic energies greater than 250 eV	electrons with kinetic energies lower than 250 eV
A	impact on plate X	impact on plate X
B	impact on plate X	impact on plate Y
C	impact on plate Y	impact on plate X
D	impact on plate Y	impact on plate Y

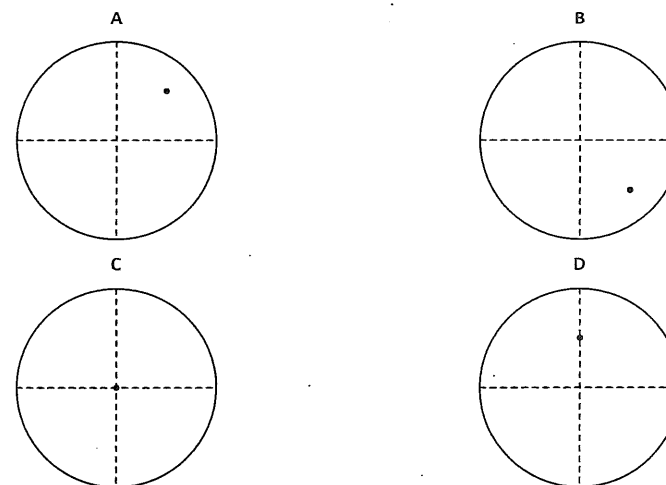
7. [EJC/H2/Prelims 2018/P1/21]

In a cathode-ray oscilloscope tube, the electron beam passes through a region where there is an electric field directed vertically downwards, and a magnetic field directed vertically upwards as shown in the diagram below.



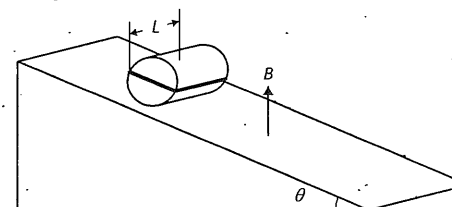
Front View of Screen

Which of the diagrams below shows a possible position of the spot on the screen when both fields are operating together?



8. [EJC/H2/Prelims 2018/P1/22]

A wooden cylinder of mass 0.250 kg and length L of 0.100 m has 10 turns of wire wrapped around it longitudinally. The cylinder is released on a plane inclined at an angle θ to the horizontal, with the plane of the coil parallel to the incline plane, as shown in the diagram below.

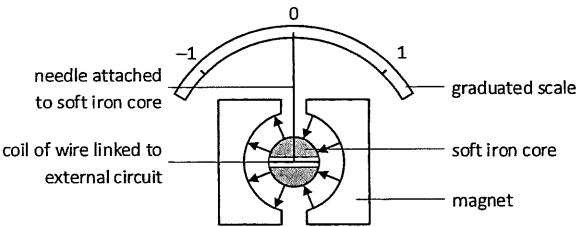


If there is a vertical uniform magnetic field of magnitude 0.500 T acting throughout the plane, what is the least current through the coil that keeps the cylinder from rolling down the plane?

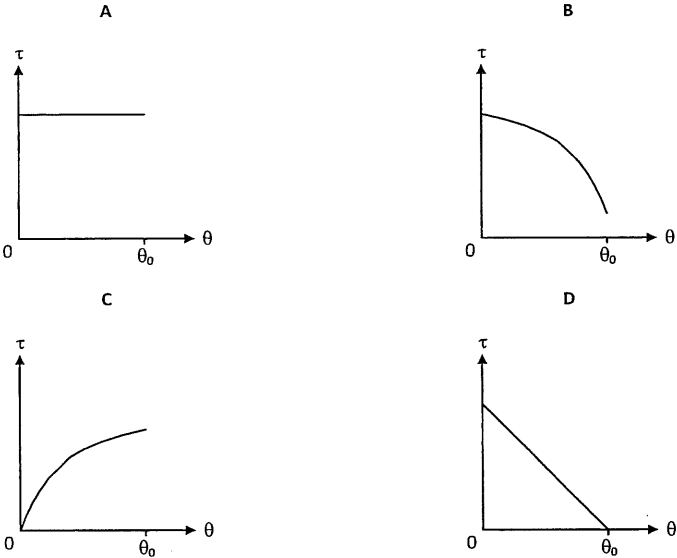
- A 1.25 A B 2.45 A C 6.77 A D 11.2 A

9. [H1/H2/Prelims 2018/P1/22]

The figure shows one of the earliest designs of a galvanometer. A coil of wire is wound around a cylindrical soft iron core which is placed between two magnets with cylindrical surfaces such that the magnetic field (indicated by the arrows) on the surface of the iron core is directed perpendicularly onto the surface and of the same magnitude across the surface of the iron core.

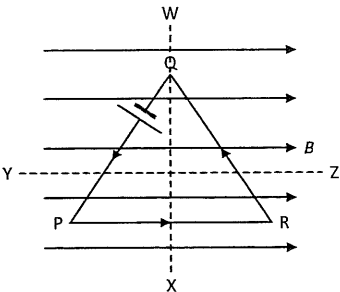


When a constant current flows in the coil, the needle will be deflected to an angle θ_0 from the zero direction. Which of the following graphs show the variation with the angular displacement θ of the needle, of the torque on the soft iron core τ , due to magnetic forces acting on the coil?



10. [JC/H2/Prelims 2018/P1/23]

A triangular piece of wire frame is placed in a magnetic field as shown below.



When a current I is supplied as shown, how does the wire frame rotate?

	Axis of rotation	Direction of movement
A	YZ	Q into page
B	YZ	Q out of page
C	WX	R into page
D	WX	R out of page

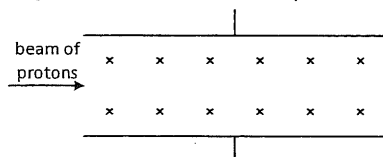
11. [JC/H2/Prelims 2018/P1/23]

An electron moves in a circular orbit in a uniform magnetic field. Which of the following statements is correct?

- A The period of the orbit is independent of the speed of the electron.
- B The momentum of the electron is dependent on its charge.
- C The radius of the orbit is directly proportional to its charge.
- D The magnetic force on the electron is dependent on the mass of the electron.

12. [JJC/H2/Prelims 2018/P1/24]

A beam of protons passes through a velocity selector with a plate separation of 2.00 cm as shown in the diagram below. The magnetic flux density B is 1.5 T and directed into the plane of the paper.

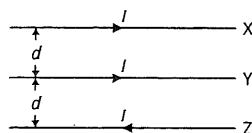


If protons travelling at $2.00 \times 10^7 \text{ m s}^{-1}$ pass through undeflected, what would be the direction and magnitude of the electric field?

	Direction	Magnitude
A	Downwards	$6.00 \times 10^5 \text{ N C}^{-1}$
B	Upwards	$6.00 \times 10^5 \text{ N C}^{-1}$
C	Downwards	$3.00 \times 10^7 \text{ N C}^{-1}$
D	Upwards	$3.00 \times 10^7 \text{ N C}^{-1}$

13. [MI/H2/Prelims 2018/P1/25]

The three parallel wires X, Y and Z carry currents of equal magnitude I in the direction shown.

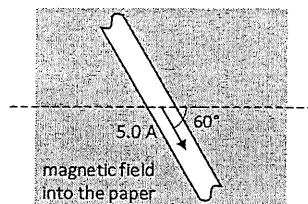


Which one of the following gives the direction of the resultant force experienced by Y due to the currents in X and Z?

- A Along Y
- B Towards Z
- C Towards X
- D Along Z

14. [MJC/H2/Prelims 2018/P1/22]

A wire of length 3.0 cm is placed in the plane of the paper, along a line 60° clockwise from the x-axis. A magnetic field of flux density 0.040 T acts into the paper. The wire carries a current of 5.0 A.



What is the magnitude of the force which the field exerts on the wire?

- A 0.0060 N
- B 0.0030 N
- C 0.0052 N
- D 0.0104 N

15. [MJC/H2/Prelims 2018/P1/23 Modified]

An electron is moving along the axis of a solenoid carrying a current.

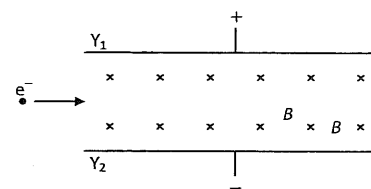
Which of the following is a correct statement about the magnetic force acting on the electron?

- A No force acts on the electron.
- B The force acts in the direction of motion.
- C The force acts opposite to the direction of motion.
- D The force causes the electron to move along a helical path.

16. [NJC/H2/Prelims 2018/P1/17]

An electron is accelerated from rest through a potential difference V (not shown in the figure). It then passes without any deflection through a region of mutually perpendicular electric and magnetic fields.

The electric field is provided by a pair of horizontal plates Y_1 and Y_2 with potential difference V and separation d . The uniform magnetic field has magnetic flux density B .

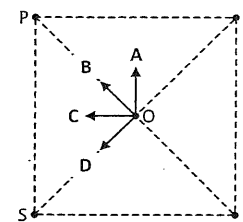


The charge to mass ratio of the electron is given by

- A $\frac{2B^2 d^2}{V}$
- B $\frac{V}{2B^2 d^2}$
- C $\frac{B^2 d^2}{2V}$
- D $\frac{2V}{B^2 d^2}$

17. [PJC/H2/Prelims 2018/P1/21]

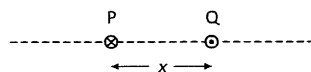
The figure below shows four long, straight current-carrying wires P, Q, R and S which are perpendicular to the plane of the paper. They pass through the corners of a square. Point O is the point of intersection of the diagonals of the square. The currents in all four wires have the same magnitude. The currents in wires P, Q and R flow into the plane of the paper while that in S flows out of the plane of the paper.



What is the direction of the resultant magnetic field at O?

18. [RVHS/H2/Prelims 2018/P1/23 Modified]

Two parallel wires P and Q are placed a distance x from each other. P carries a current of I into the paper while Q carries a current of $2I$ out of the paper.



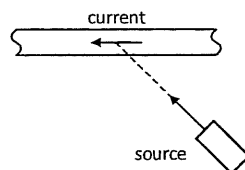
A third current-carrying wire R is placed along the dotted line, parallel to wires P and Q, and experiences zero resultant force.

Which of the following is a possible combination of the position of wire R, and the direction and magnitude of current in R?

	position of R	direction of current in R	magnitude of current in R
A	x to the left of P	out of paper	$2I$
B	$0.5x$ to the left of P	into paper	I
C	$2x$ to the left of P	out of paper	I
D	x to the right of Q	out of paper	I

19. [RVHS/H2/Prelims 2018/P1/24]

The diagram shows a long horizontal wire carrying a steady current. A source fires a beam of electrons towards the wire at an angle of 45° .

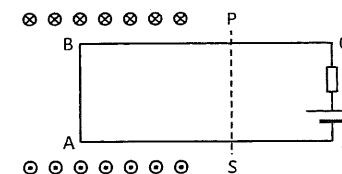


As the beam approaches the wire, the electrons experience a magnetic force due to the field of the current. What is the initial direction of this force?

- A In a direction parallel to the wire
- B In a vertical direction towards the wire
- C In a direction at 90° to the original path towards the wire
- D In a direction at 90° to the original path away from the wire

20. [SRJC/H2/Prelims 2018/P1/24]

The figure shows the top view of a current balance where the light, rigid rectangular wire loop ABQR of negligible resistance is pivoted at PS. It is connected in series with a battery of e.m.f. 2.0 V and a resistor of 2.5Ω . The total mass of the battery and the resistor is 30 g. Part of the wire loop is placed inside a solenoid.

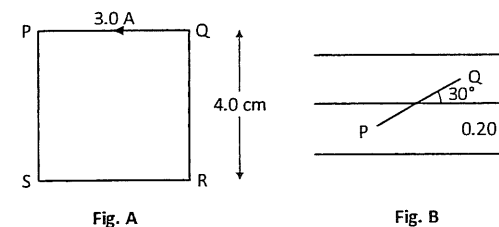


Given that the side AB is 6.0 cm and $SR = \frac{2}{3}AR$, what is the magnitude of the magnetic flux density in the solenoid?

- A 3.7 T
- B 4.1 T
- C 37 T
- D 41 T

21. [TPJC/H2/Prelims 2018/P1/24]

Fig. A shows a vertical plane square coil of 50 turns, carrying a current of 3.0 A. The length of each side of the coil is 4.0 cm. Fig. B shows a view of this coil from above within a horizontal magnetic field of flux density 0.20 T.

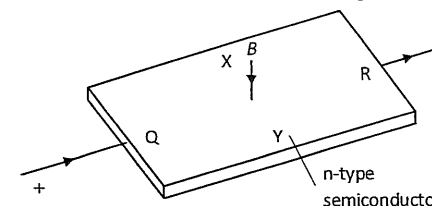


What is the magnitude of the resultant moment on the square coil?

- A 0 N m
- B 0.024 N m
- C 0.042 N m
- D 0.048 N m

22. [TPJC/H2/Prelims 2018/P1/25]

In n-type semiconductors, majority of mobile charge carriers are conduction electrons. A thin slab of n-type semiconductor is connected at its ends to a battery causing a current I through it as shown. A uniform magnetic field B is applied vertically downwards over the surface of the slab causing an electric field to be induced in the slab.

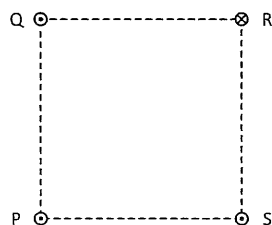


Which of the options give the correct direction of the induced electric field?

- A from Q to R
- B from R to Q
- C from X to Y
- D from Y to X

23. [TPJC/H2/Prelims 2018/P1/26 Modified]

Four long straight current-carrying wires are arranged at the corners of a square PQRS as shown. A current of I flows out of the page in wires P, Q and S while the current in wire R is into the page in wire R.

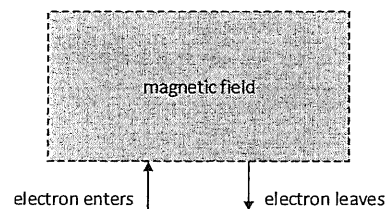


If the wire at P experiences zero resultant force, what is the amount of current flowing along the wire at R?

- A $\frac{I}{\sqrt{2}}$ B $\frac{I}{2}$ C $\sqrt{2}I$ D $2I$

24. [VJC/H2/Prelims 2018/P1/23 Modified]

An electron travelling at a speed of $5.0 \times 10^7 \text{ m s}^{-1}$ enters perpendicularly into a rectangular region of uniform magnetic field of strength 3.0 T , as shown. The field is perpendicular to the page.



How much time does the electron spend inside the magnetic field?

- A $6.0 \times 10^{-31} \text{ s}$ B $1.2 \times 10^{-30} \text{ s}$ C $6.0 \times 10^{-12} \text{ s}$ D $1.2 \times 10^{-11} \text{ s}$

25. [VJC/H2/Prelims 2018/P1/24]

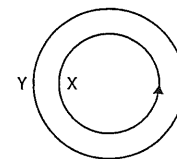
A straight wire XY 2.0 cm long carrying a current of 1.5 A is placed at the centre of a solenoid 1.2 m long with 20000 turns of wire and carrying a current of 3.0 A , so that the wire XY is perpendicular to the magnetic field produced by the solenoid.

What is the magnetic force acting on the wire XY?

- A $6.0 \times 10^{-4} \text{ N}$ B $7.2 \times 10^{-4} \text{ N}$ C $1.9 \times 10^{-3} \text{ N}$ D $2.3 \times 10^{-3} \text{ N}$

26. [YJC/H2/Prelims 2018/P1/25]

Two flat circular coils, X and Y, each with 100 turns, are arranged as shown. X has radius 0.12 m and carries a current of 1.5 A in the anticlockwise direction, Y has radius 0.18 m and carries a current of 1.0 A in the clockwise direction. Both coils are arranged such that their centres coincide.

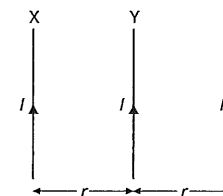


What is the magnitude and direction of the total magnetic flux density at the centre of the coils?

- A $347 \mu_0$ into the page
B $347 \mu_0$ out of the page
C $903 \mu_0$ into the page
D $903 \mu_0$ out of the page

27. [YJC/H2/Prelims 2018/P1/26]

The figure below shows three parallel wires X, Y and Z which carry currents I of equal magnitude in the directions shown.



The resultant force experienced by X due to the currents in Y and Z is

- A zero.
B away from Y.
C towards Y.
D along Y.

II. Structured Questions (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P3/6 Part Modified]

- b. i. A long positively charged rod is placed at a distance from a stationary $-2.0 \mu\text{C}$ point charge as shown in Fig. 1.1.

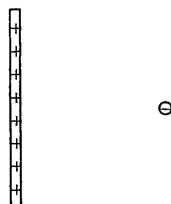


Fig. 1.1

Sketch the electric field pattern between the rod and the point charge.

[1]

- ii. A current I is passed through the rod and the charge moves at a speed of 1.5 cm s^{-1} as shown in Fig. 1.2. The charge experiences a magnetic flux density of 10 mT .

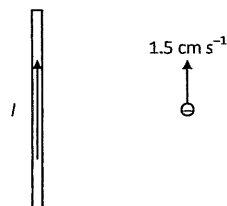


Fig. 1.2

- Calculate the magnitude of the force experienced by the charge due to the magnetic field of the rod. [2]
 - The negative charge moves in a curved path to the right. Explain this observation. [2]
 - State and explain a modification that can be made to Fig. 1.2 such that the negative charge can be made to move parallel to the rod when the current I is still passing through the rod. [2]
- iii. In a second experiment, the rod carrying a current of 4.0 mA is placed next to a horizontal flat coil of wire which is fixed in its position. The centre of the coil, P, is at a perpendicular distance of 20 cm from the rod as shown in Fig. 1.3.

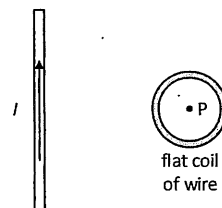


Fig. 1.3

- Calculate the magnitude of the magnetic flux density at P. [2]
- A very small current now flows in the flat coil of wire in an anticlockwise direction. Without any further calculation, state and explain if the magnetic flux density at P will change compared to that obtained in b.iii.1. [2]
- State and explain the direction of the resultant force acting on the coil. [2]

[Total: 13]

2. [AJC/H2/Prelims 2018/P3/7 Part]*

- c. A large horseshoe magnet produces a uniform magnetic field of flux density B between its poles. Outside the region of the poles, the flux density is zero. The magnet is placed on a top-pan balance and the wire XY is situated between its poles, as shown in Fig. 2.1.

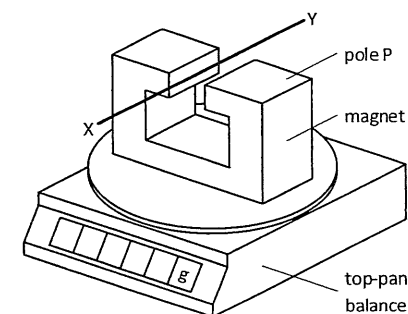


Fig. 2.1

The wire XY is horizontal and normal to the magnetic field. The length of wire between the poles is 4.4 cm . A direct current of magnitude 2.6 A is passed through the wire in the direction from X to Y. The reading on the top-pan balance increases by 2.3 g .

- State and explain the polarity of the pole P of the magnet. [3]
- Calculate the magnetic flux density between the poles. [2]
- A low frequency alternating current is now passed through the wire. The root-mean-square (r.m.s.) value of the current is 2.6 A . Describe quantitatively the variation of the reading seen on the balance. [2]

[Total: 7]

3. [CJC/H2/Prelims 2018/P2/6]

- a. Define magnetic flux density. [2]
- b. Fig. 3.1 shows a long, straight, vertical wire WX, carrying a current of 9.0 A downwards. A second long, straight wire YZ is placed horizontally, and carries a current of 12.0 A in the direction shown.

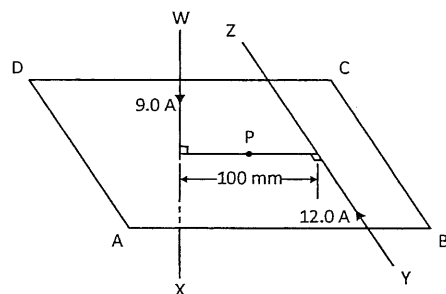


Fig. 3.1

ABCD is a horizontal, rectangular tabletop. The wire YZ is parallel to the side BC of this table, and the wire WX passes through a small hole in the table. The perpendicular distance between the wires is 100 mm. P is the point 50 mm from YZ along the perpendicular between the wires.

- i. Determine the magnitude of the magnetic flux density at the point P due to WX only. [2]
- iii. Determine the magnitude of the resultant magnetic flux density at the point P. [3]

[Total: 7]

4. [CJC/H2/Prelims 2018/P3/5]

- a. A particle of mass m , carrying a negative charge $-q$ and travelling at speed v , enters a region of uniform magnetic field of flux density B directed at right angles to the motion of the particle as shown in Fig. 4.1.

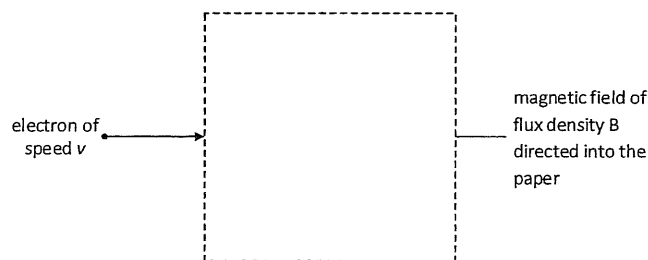


Fig. 4.1

- i. State the expression for the magnitude of the force F acting on the particle and the direction of the force. [2]
- ii. Explain why the path of the electron is circular. [3]

- b. The diagram in Fig. 4.2 shows a type of cathode ray tube containing a small quantity of gas. Electrons from a hot cathode emerge from a small hole in the conical shaped anode, and the path subsequently followed is made visible by the gas in the tube.

The accelerating voltage is 5.0 kV.

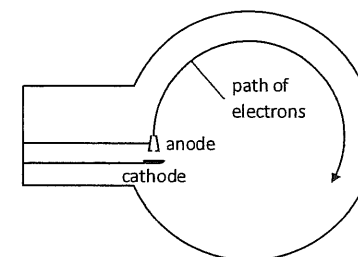


Fig. 4.2

- i. Calculate the speed of the electrons as they emerge from the anode. [2]
- ii. The apparatus is situated in a uniform magnetic field acting into the plane of the paper.
- Calculate the radius of the circular path for a flux density of $2.0 \times 10^{-3} \text{ T}$. [2]
 - Suggest how the gas in the tube might make the path of the electrons visible. [1]

[Total: 10]

5. [DHS/H2/Prelims 2018/P2/6]

- a. Two identical wires A and B are placed parallel to each other as shown in Fig. 5.1. Both wires carry a current of 90.0 A towards the left.

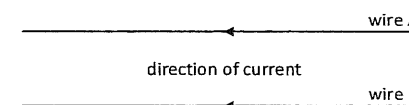


Fig. 5.1

Explain why both wires experience a force. [2]

- b. Fig. 5.2 shows only the wire A carrying a current of 90.0 A towards the left. Point P lies in the plane of the paper containing wire A, at a distance 50.0 cm directly above wire A. At P, a proton is travelling directly towards the wire with a speed $v = 1.00 \times 10^3 \text{ m s}^{-1}$. Ignore the effects of the Earth's magnetic field.

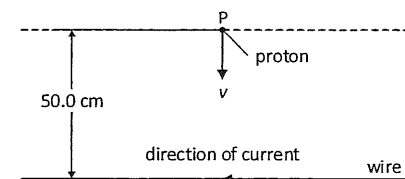


Fig. 5.2

- i. Calculate the radius of the path of the proton when it is at P. [3]
- ii. Explain how the path of the proton is affected by the magnetic field produced by the current in wire A as it moves in the region between P and wire A. [3]

[Total: 8]

6. [EJC/H2/Prelims 2018/P2/3]

- a. Define the tesla. [1]
- b. Charged particles, of speed 4500 m s^{-1} and mass $2.66 \times 10^{-26} \text{ kg}$, are travelling in a narrow beam in a vacuum as shown in Fig. 6.1.

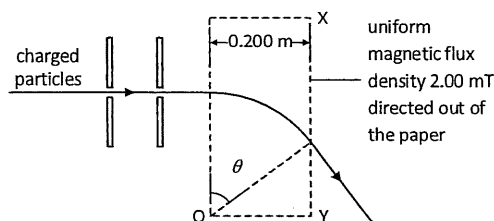


Fig. 6.1

The charged particles enter a region of uniform magnetic flux density which is 0.200 m wide. The direction of the magnetic flux density is pointed out of the paper.

- Using Newton's Law of motion, state and explain the speed of the particles as they exit the uniform magnetic field. [2]
- Given that the magnitude of the charge on the particles is e , calculate the radius of the circular motion of the charged particles in the uniform magnetic field. [2]
- Hence, calculate the angular velocity of the charged particles in the uniform magnetic field. [1]
- Point O is the centre of the circular motion of the charged particles as shown in Fig. 6.1. Show that the angle θ is 0.564 rad. [2]
- In another experiment, similar charged particles are now fired into the magnetic field in Fig. 6.1 with different momentum.
 - Determine the maximum momentum of particles such that the particles will not exit the magnetic field through XY. [2]
 - Show that the time taken for the particle of mass m and charge q in b.v.1. to complete a semi-circle within a region of magnetic flux density B is given by $\frac{\pi m}{Bq}$. [2]

[Total: 12]

7. [JC/H2/Prelims 2018/P3/8 Modified]

- a. Define the tesla. [2]
- b. Fig. 7.1 shows an arrangement used to accelerate an initially stationary alpha particle and make it travel in a uniform magnetic field.

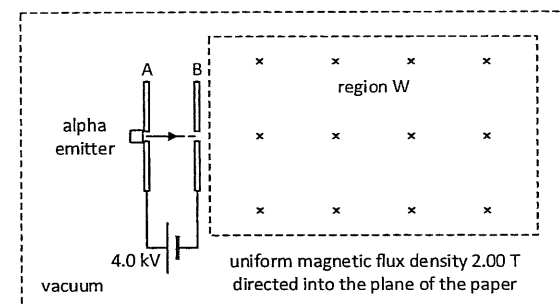


Fig. 7.1

- On Fig. 7.1, draw a possible trajectory of the alpha particle in the uniform magnetic field. [1]
 - Explain whether the force experienced by the alpha particle due to the magnetic field changes its kinetic energy. [3]
 - Show that the alpha particle will attain a speed of $6.2 \times 10^5 \text{ m s}^{-1}$ when it reaches the slit opening of plate B. [1]
 - Determine the electric field that needs to be applied in Region W for the alpha particle to pass through the uniform magnetic field undeflected. [3]
- c. Suppose that the arrangement in Fig. 7.1 is now modified so that the alpha particle enters the uniform magnetic field at an angle of 30° to the horizontal as shown in Fig. 7.2 below.

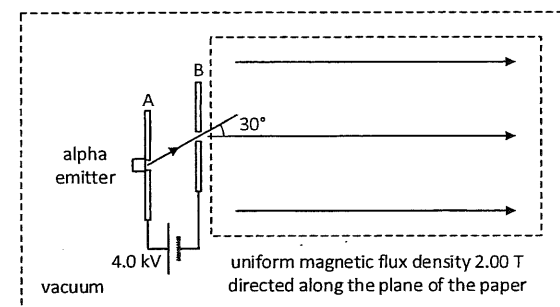


Fig. 7.2

The path of the alpha particle can be described as a helix as shown in Fig. 7.3.

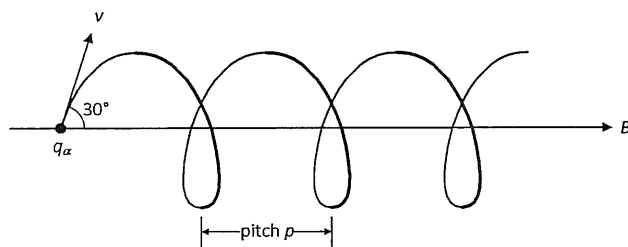


Fig. 7.3 (not to scale)

- i. Calculate the radius r of the helical path. [2]
- ii. Show that the period, T of the helical path can be expressed as

$$T = \frac{2\pi m_\alpha}{q_\alpha B}$$

where m_α and q_α are the mass and charge of the alpha particle respectively. [1]

- iii. Calculate the pitch p . [3]
- iv. Describe and explain how the helical path will change if a positron ${}^0_{+1}\text{e}$ (a particle with the same mass as an electron but with opposite charge) with the same initial velocity was to be used in the experiment instead. You may quote relevant equations to substantiate your answer. [4]

[Total: 20]

8. [JJC/H2/Prelims 2018/P3/9 Part]

- a. Define magnetic flux density. [2]
- b. Fig. 8.1 below shows a current carried by a square coil GHJK of 500 turns and side 5.0 cm. The coil is suspended vertically in a uniform horizontal magnetic field of flux density $B = 2.0 \times 10^{-4} \text{ T}$. The plane of the coil is at angle $\theta = 35^\circ$ to B .

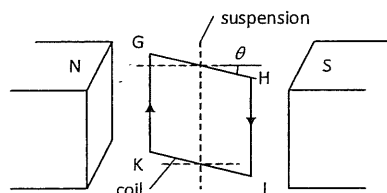


Fig. 8.1

- i. Indicate in Fig. 8.1, the direction of the magnetic force acting on the side JK of the coil. [1]
- ii. The current in coil GHJK is 0.40 A. Calculate the magnetic force acting on side JK of the coil. [2]
- iii. Calculate the torque acting on the coil GHJK. [3]

[Total: 8]

9. [MI/H2/Prelims 2018/P3/S Modified]

- a. i. Fig. 9.1 shows the cross section of a wire XY which carrying a constant current flowing out of the plane of the paper at right angles. Sketch the magnetic flux pattern due to the current. [2]



Fig. 9.1

- ii. The current-carrying wire XY in a.i. is placed in the region between the poles of a strong magnet. Sketch in Fig. 9.2 the resultant magnetic flux pattern in the region between the poles of a magnet. [3]

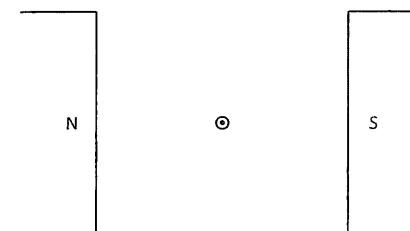


Fig. 9.2

- b. The setup in a. is now placed on a top-pan balance as shown in Fig. 9.3 and Fig. 9.2 shows the front view. Outside the region between the poles of the strong magnet, the flux density can be considered as zero. When there is no current passing through XY, the top-pan balance is tared.

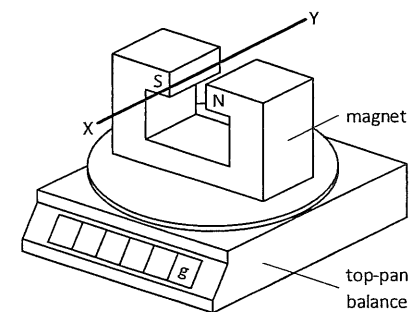


Fig. 9.3

A direct current is now passed through the wire in the direction from Y to X. State and explain the observation in the reading on the top-pan balance.

[2]

[Total: 7]

10. [MJC/H2/Prelims 2018/P3/8 Part Modified]

- b. Fig. 8.2 shows an electron entering a region between two oppositely charged parallel metal plates. The plates have length 5.1 cm.
 The electric field in the region between the plates is uniform and is zero outside this region.
 The original direction of motion of the electron is normal to the electric field.
 The original speed of the electron is $v = 1.70 \times 10^7 \text{ m s}^{-1}$.
 The electric field strength between the plates E is 4000 V m^{-1} .
 The electron exits the plates at an angle θ to the horizontal.

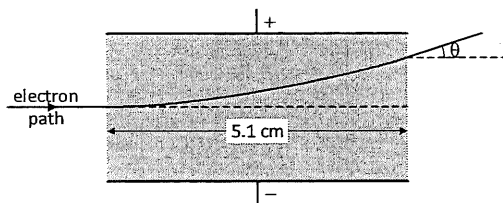


Fig. 10.1

- Show that the acceleration of the electron inside the electric field is $7.03 \times 10^{14} \text{ m s}^{-2}$. [1]
 - Calculate the magnitude of the final velocity of the electron, and the angle θ . [4]
 - A proton is projected with the same initial velocity along the same line. Without detailed calculation, draw the path that the proton takes on Fig. 10.2. Explain your answer. [4]
- c. Fig. 10.2 shows a uniform magnetic field B denoted by the shaded area. An electron moves into the field at the same speed v as in b., and is deflected from its original path. The original direction of motion of the electron is normal to the magnetic field.

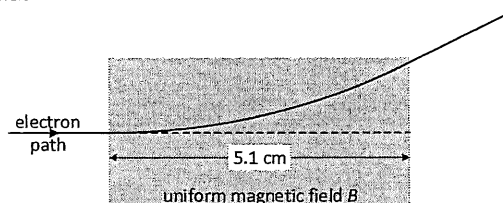


Fig. 10.2

State the difference between the shape of the path taken by the electron in the magnetic field, and the shape of the path taken by the electron in the electric field described in b. Explain this difference. [3]

[Total: 12]

11. [NJC/H2/Prelims 2018/P3/4]

Fig. 11.1 shows a mass spectrometer comprising a velocity selector (Stage 1) and an ion separator (Stage 2). A stream of alpha particles is sent into the mass spectrometer.

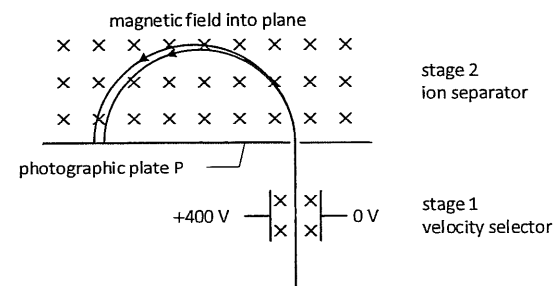


Fig. 11.1

a. Stage 1

The alpha particles have a range of speeds enter Stage 1 which isolates particles with a specific speed v . The plates producing the electric field have a separation of 1.0 cm. A uniform magnetic field is applied between the plates.

- On the diagram below, draw arrows to represent the electric force F_E and the magnetic force F_B . [2]



- Describe and explain the path of alpha particles with speed less than v . [3]

b. Stage 2

After Stage 1, alpha particles of speed $1.00 \times 10^5 \text{ m s}^{-1}$ are sent into stage 2. The alpha particles are travelling at right angles to the uniform magnetic field and are detected using a photographic plate P placed at the appropriate position.

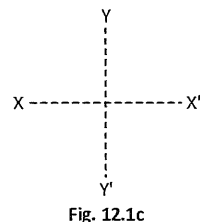
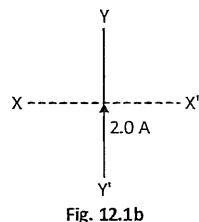
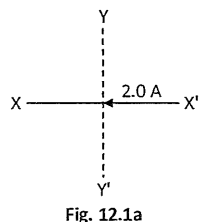
Determine the distance between the point of entry and the point of impact of the alpha particles with the photographic plate when the magnetic flux density 0.050 T is used. [2]

- A similar spectrometer can be used with beta particle. Beta particles typically travel more than 100 times faster than the alpha particles. State and explain the necessary modifications to the spectrometer for it to work with beta particles. [4]

[Total: 11]

12. [NYJC/H2/Prelims 2018/P2/7 Modified]

- a. Define magnetic flux density. [1]
- b. A 1.0 m long wire carrying a current of 2.0 A is placed in a magnetic field. When it is aligned with XX' (Fig. 12.1a), it experiences a force of 0.50 N directed into the page. When it is aligned with YY' (Fig. 12.1b), it experiences a force of 0.80 N directed out of the page.



- i. Calculate the magnetic flux density of the field present. [3]
- ii. Determine the position of the wire such that it experiences a force of the maximum magnitude. Draw accurately the position of the wire in Fig. 12.1c. [2]
- c. The current-carrying wire in b. has a magnetic field of its own. Show, by calculation, that it is impossible to find a point near the wire such that its magnetic flux density at that point has the same magnitude as the field present. [2]

[Total: 8]

13. [PJC/H2/Prelims 2018/P2/6]

- a. i. Write down the equation defining *magnetic flux density* in terms of the force F acting on a long, straight conductor of length L carrying a current I at an angle θ to the field. [1]
- ii. Draw a diagram to illustrate the direction of the force relative to the current and magnetic field. [1]
- b. Fig. 13.1 shows a small square coil of N turns and sides of length L . It is mounted so that it can pivot freely through the centre of the coil, about a horizontal axis PQ parallel to one pair of sides of the coil.

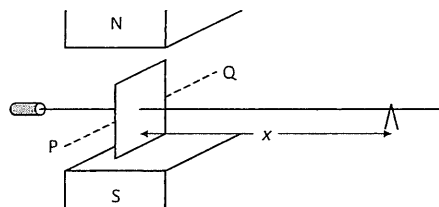


Fig. 13.1

The coil is placed between the poles of a magnet which produces a uniform magnetic flux density B . The coil is maintained in a vertical plane by moving a rider of mass M along a horizontal beam attached to the coil. When a current I flow through the coil, equilibrium is restored by adjusting x , the distance of the rider from the coil.

- i. Starting from the definition of magnetic flux density, show that B is given by the expression $B = \frac{Mgx}{IL^2N}$. [3]
- ii. Current I is supplied by a battery of constant e.m.f. E and negligible internal resistance. Discuss the effect on x if the coil is replaced by one wound with wire of the same material and diameter but forming a coil of N turns with sides of length $L/2$. [3]

[Total: 8]

14. [RVHS/H2/Prelims 2018/P3/4 Modified]

A cyclotron uses a magnetic field to maintain alpha particles in nearly circular paths. Electromagnets with ferrous cores provide the required strong magnetic field as shown in Fig. 14.1.

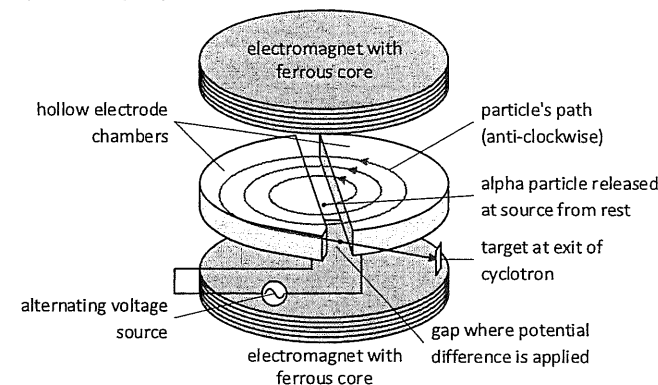


Fig. 14.1

The alpha particles, originating from a source at the centre, released from rest, move within two D-shaped hollow electrode chambers called *dees*, separated by a very narrow gap.

Each time the alpha particles pass into the gap between the *dees*, they are accelerated by a potential difference between the *dees*, subsequently increasing the radius of curvature of their path. After many revolutions, the alpha particles acquire high speeds and reach the outer edge of the cyclotron.

- a. State what is meant by a *magnetic field*. [1]
- b. On Fig. 14.1, indicate the direction of magnetic flux density provided by the electromagnets. [1]
- c. i. By considering the path taken by the alpha particles and forces acting on them in the cyclotron, derive an expression for the radius of curvature r of the path. Express your answer in terms of mass of alpha particles, charge, linear speed v , and magnetic flux density B . Charge should be in terms of elementary charge e and mass in terms of unified atomic mass constant u . Show your working clearly. [2]
- ii. Hence, explain why the radius increases each time after the alpha particles pass between the *dees*. [1]
- d. The magnetic flux density in the *dees* provided by the electromagnets is equivalent to that at the centre of a solenoid of length 0.39 m, 3100 turns, and a current of 200 A flowing in it. Show that the magnetic flux density in the *dees* is 2.0 T. [1]
- e. The potential difference V between *dees* varies periodically as shown in Fig. 14.2.

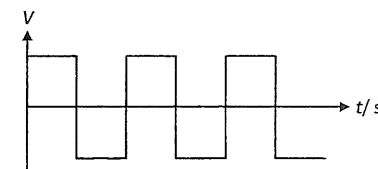


Fig. 14.2

The alpha particles reach an energy of 16 MeV after accelerating within the cyclotron for a duration of 0.52 ms and exiting the cyclotron at a radius of 0.29 m. Determine the potential difference between the *dees* at each crossing. [4]

[Total: 10]

15. [TPJC/H2/Prelims 2018/P3/5 Modified]

A *cyclotron* is a device that accelerates charged particles to high energies. The cyclotron uses a magnetic field to keep charged particles moving in circles. The charged particles move within two D-shaped hollow metal conductors called *dees*, which are separated by a very narrow gap.

An alternating square-wave potential difference is applied between the dees, resulting in a region of electric field at the gap. Each time the charged particles pass into the gap between the dees, they are accelerated by the potential difference across the gap. This increases their speed and also increases the radius of curvature of their path. After many revolutions, the charged particles acquire high kinetic energies and exit the outer edge of the cyclotron.

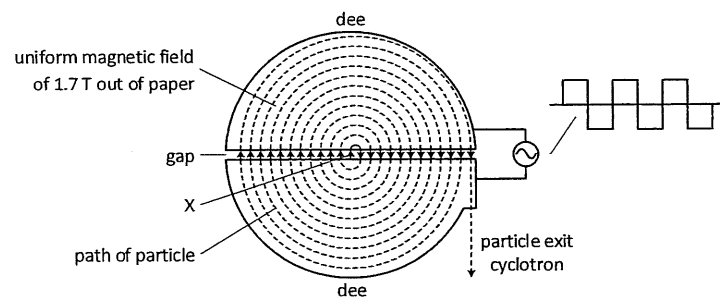


Fig. 15.1

A particle enters the centre X of the cyclotron, as shown in Fig. 15.1, at a *low speed*, and travels in circular paths under the influence of a uniform magnetic field of flux density 1.7 T, which is acting perpendicularly out the page.

- With reference to Fig 15.1, state whether the particle is an electron or a proton. [1]
- The frequency of the alternating potential difference applied between the dees has to be set accurately to be equal to the frequency of revolutions of the charged particle. Explain what will happen if the frequency is not set accurately. [1]
 - Show that the frequency needed for the alternating potential difference is 2.6×10^7 Hz. [3]
- Just before exiting the cyclotron, the particle was traveling in a path of radius 0.25 m and had just completed 10000 revolutions. Determine
 - the kinetic energy of the particle just before it exits the cyclotron. [3]
 - the amplitude of the alternating potential difference applied across the dees. [2]

[Total: 10]

16. [VJC/H2/Prelims 2018/P3/4 Modified]

Fig. 16.1 shows a light rectangular wire frame ABCD of sides 15.0 cm and 6.0 cm acting as a current balance. The wire frame is pivoted such that the pivot is 8.0 cm from AB. A 2.0 g mass is attached to CD. Side AB is entirely immersed in a uniform horizontal magnetic field B which is perpendicular to AB.

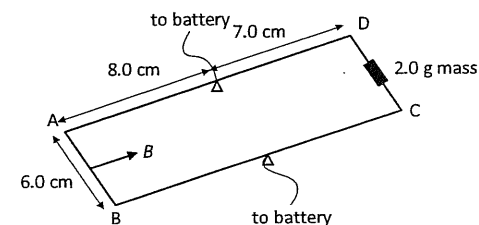


Fig. 16.1

A current of 3.0 A flows through the wire frame, and the frame is found to come to a rest in the horizontal plane.

- State the direction (A to B or B to A) of the current. [1]
- Calculate the magnetic field B . [3]
- Explain why such a current balance cannot be used to measure the strength of a magnetic field that is not horizontal. [3]

[Total: 7]

17. [YJC/H2/Prelims 2018/P3/6 Part Modified]

Fig. 17.1 shows the main components of a mass spectrometer which are an ion chamber and a velocity selector. Positive ions emerging from the ion chamber pass through the velocity selector. A uniform electric field and a uniform magnetic field (indicated by the shaded area) are applied in region R. While only a uniform magnetic field (indicated by the shaded area) is applied in region S.

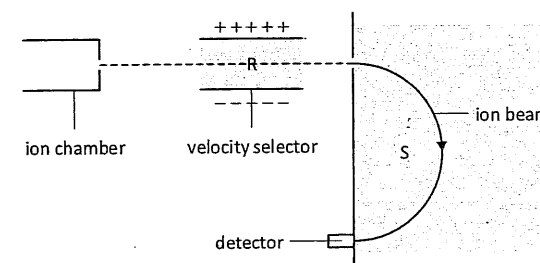


Fig. 17.1

- Draw the electric field lines between the plates in the region R of the velocity selector in Fig. 17.1. [2]
- Suggest the purpose of the two slits. [1]
- In the regions R and S, uniform magnetic fields are applied so that the beam of positive ions is undeflected in R and deflected into a circular path in S.
 - State the direction of the magnetic fields in regions R and S. [2]
 - Explain why positive ions entering region S have the same speed regardless of their mass. [3]
 - State and explain what happens to the slower moving ions that enters region R. [2]

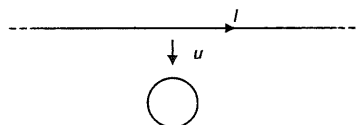
[Total: 10]

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [AJC/H2/Prelims 2018/P1/23]

A long straight wire is in the plane of a flat short-circuited coil. The straight wire carries a constant current I as shown and is moved at a constant speed u towards the flat coil.

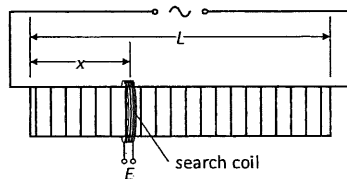


Which statement describing the current in the flat coil is true?

- A The current is always zero.
- B The current is always constant.
- C The current is decreasing.
- D The current is increasing.

2. [CJC/H2/Prelims 2018/P1/26]

In the diagram below, a solenoid of length l which is closely and uniformly wound, carries an alternating current of constant amplitude. A search coil (which is a coil consisting of a few turns of wires) is placed in different positions along the solenoid.



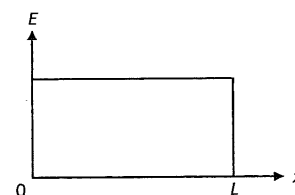
Which of the graphs shows how the amplitude of the induced e.m.f. E in the search coil varies with its position?

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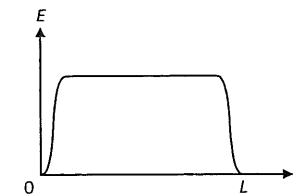
Revision Package (H2 Physics)

18 – Electromagnetic Induction

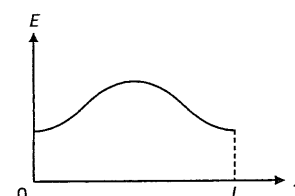
A



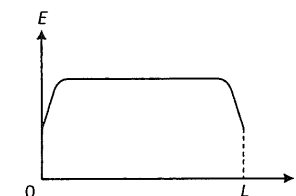
B



C



D



3. [DHS/H2/Prelims 2018/P1/25 Modified]

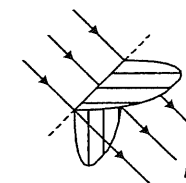
A coil of 160 turns and area 0.20 m^2 is placed with its axis parallel to a magnetic field in the x -direction. The magnetic flux density changes from 0.40 T in the positive x -direction to 0.40 T in the negative x -direction in 2.0 s at a constant rate.

If the resistance of the coil is 16Ω and both ends of the coils are connected, what is the rate of energy generated in the coil?

- A 5 W
- B 10 W
- C 13 W
- D 20 W

4. [EJC/H2/Prelims 2018/P1/23]

A closed circular loop of wire has a radius of 3.7 cm . It is bent along a diameter such that the two halves are perpendicular to each other. A uniform magnetic flux density of $B = 76 \text{ mT}$ is directed perpendicular to the fold diameter and makes equal angle of 45° with the planes of the semicircle.



If the magnetic flux density B is reduced to zero at a uniform rate during a time interval of 4.5 ms , what is the magnitude of the induced e.m.f. in the loop?

- A 0.026 V
- B 0.051 V
- C 0.073 V
- D 0.098 V